Advanced VLSI Design - Lab 4

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1 Layout and Schematic of Inverter 1

The schematic of the NMOS_PMOS inverter began by connecting the input to the gate of both MOSFETS and tying their drains to the output. The source of the NMOS was tied to ground while the source of the PMOS was tied to VDD. Due to electron mobility in PMOS transistors, it is a general rule of thumb that the PMOS is larger than the NMOS; thus, the transistor sizes are as follows: PMOS width = 12, length = 6; NMOS width = 6, length = 6. A corresponding icon of the inverter was created using the artwork library in the icon editor.

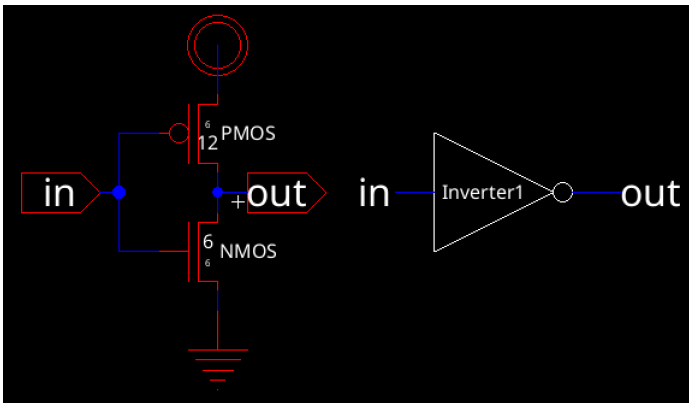


Figure 1: Schematic of Inverter 1 with Corresponding Icon

Making the layout for the inverter followed standard design philosphy that has been used throughout the course of these labs. The design began by placing an NMOS and PMOS transistor in the layout editor, followed by nACTs tied to the NMOS and pACTs tied to the PMOS. The PMOS and NMOS was sized according to the layout. An n-well was placed above the PMOS while a p-well was placed below the NMOS, representing the bodies of the transistors. To match the width of the transistors, the bodies were given a width of 26 and length of 6. A metal-1 to polysilicon-1 node was placed between the transistors and given the export name *in*, while a metal-1 pin was placed on the opposite end and given the export name *out*. Initially, there were problems matching wiring sizing between the polysilicon gates; however, after adjusting the physical parameters of the layout, the correct metal sizing could be used.

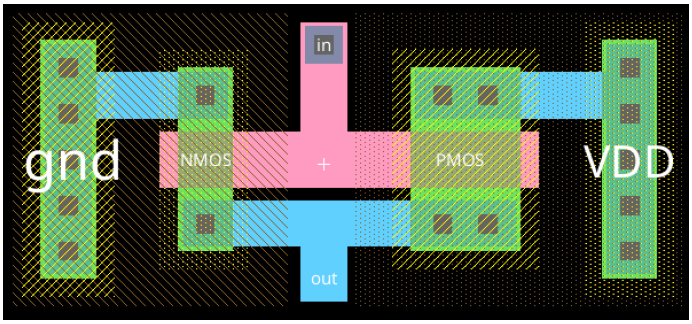


Figure 2: Layout of Inverter 1

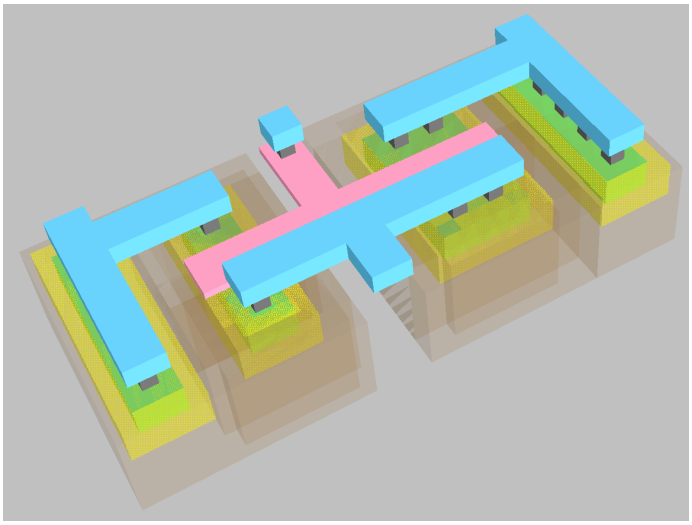


Figure 3: 3D Render of Inverter 1 Layout

2 Layout and Schematic of Inverter 2

The primary distinction between this inverter and inverter 1 is that the width has been increased by a factor of four for each MOSFET. The schametic is identical to inverter 1, but factors in this change.

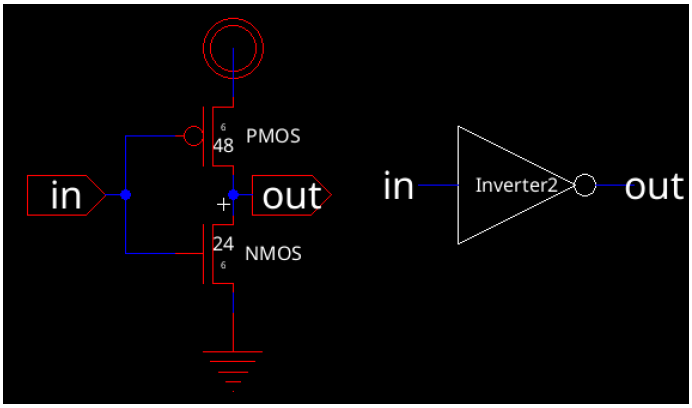


Figure 4: Schematic of Inverter 2 with Corresponding Icon

The layout of inverter 2 followed a similar approach to inverter 1; however, instead of a single 48 by 6 PMOS and 24 by 6 NMOS, four individual MOSFETS of each with identical sizing to inverter 1 were laid out and tied in parallel.

Connecting metals was identical to inverter 1, with the gates of the PMOS and NMOS tied to the input, drains tied to the output, source of the NMOS tied to ground, and source of the PMOS tied to vdd. The same desing parameters were applied to inverter 2 as they were with metal 1, where components were spaced farther apart to account for metal wiring.

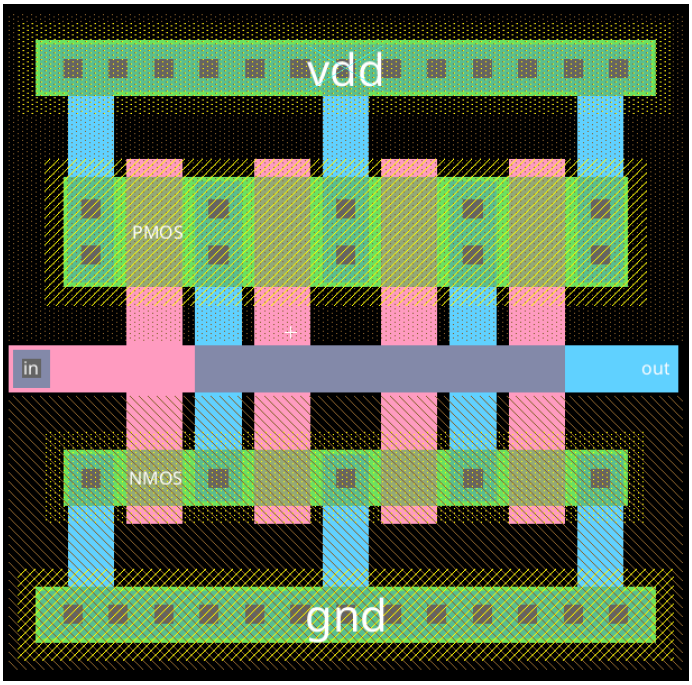


Figure 5: Layout of Inverter 2. The figure demonstrates the layout of four NMOS and PMOS transistors in parallel, designated by four poly-1 gate fingers.

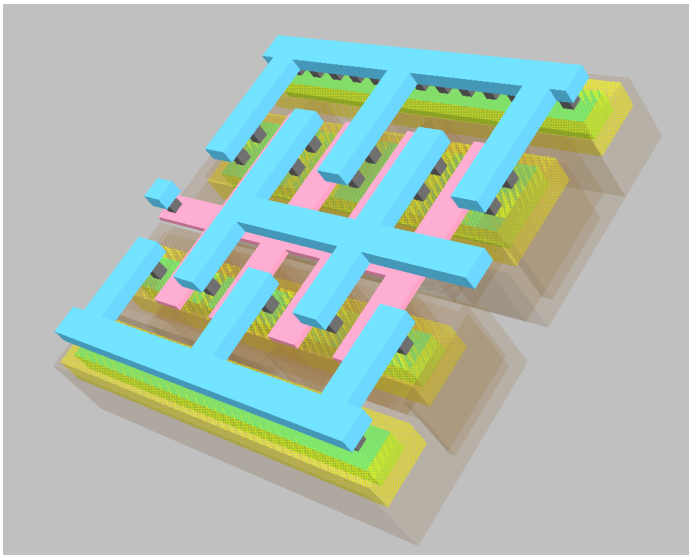


Figure 6: 3D Render of Inverter 2 Layout

3 Testing the Inverters Through LTSpice

A new schematic was created and both inverter icons were placed in the window. The two inverters were then tied together at the input via a **vin** node, and each designated their own output: **vout1** and **vout2**. The following LTspice code was written to run the simulation:

```
vdd vdd 0 DC 5
vin vin 0 DC 0
.dc vin 0 5 1m
.include *C5_models.txt
```

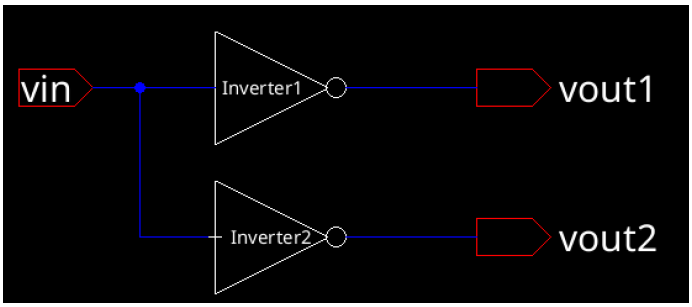


Figure 7: Schematic of Test Inverter 1

The following figure demonstrates the input voltage and output results of **vout1** and **vout2**.

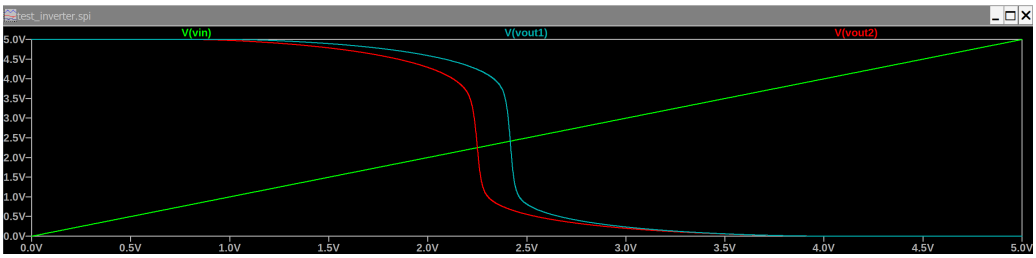


Figure 8: Simulation Results of Test Inverter 1

The second simulation that was run utilized an identical schematic as with the inclusion of a pull-down capacitor at the output of each inverter. The following LTspice code was written to run the simulation:

```
vdd vdd 0 DC 5
vin vin 0 pulse(0v 5v 5n 1n 1n 12n 25n)
.step param x list 100f 1p 10p 100p
.trans 0 25n 0 100p
.include *C5_models.txt
```

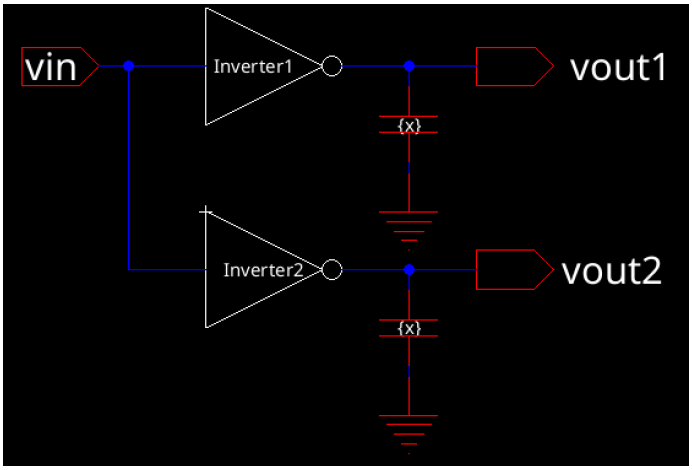


Figure 9: Schematic of Test Inverter 2

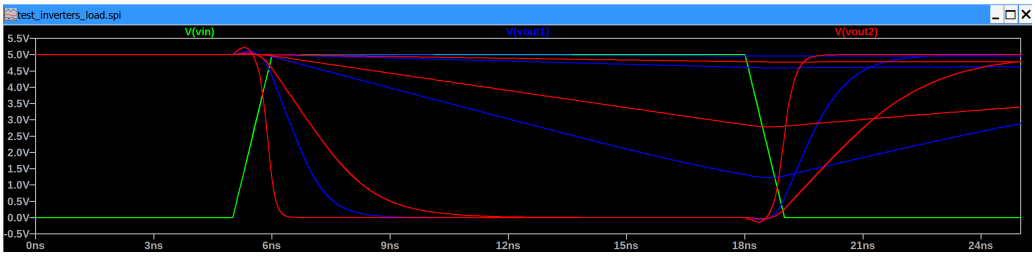


Figure 10: Simulation Results of Test Inverter with Capacitor Load

3.1 Simulation Analysis

3.2 Simulation 1

Simulation 1 uses a simple 0 to 5 volt ramp at the input to evaluate the output of both inverters and can be seen in Figure 8. The output of inverter 2 crosses the input voltage at approximately 2.25 volts, while the output of inverter 1 cross the input voltage at approximately 2.4 volts. Ideally, four MOSFETs in parallel should have near-identical results to a single inverter; however, there are likely discrepancies in the NMOS-to-PMOS strength ratio of inverter 2 that are contributing to a lower voltage intersection.

3.3 Simulation 2

Figure 10 demonstrates the results of the test inverter with the following output capacitor loads: *100 fF, 1 pF, 10 pF, 100 pF*. What the graph displays is the delay propagation of the inverted signal through these different capacitor loads, and we can use the equation $t_d = 0.35RC$ to understand how this works. At 100 fF, the rising edge and falling edge of the output is relatively steep as this

small capacitance contributes to a shorter propagation delay. As the load capacitance increases, there is a much smoother inversion of the output as the propagation delay is increased.

4 Documentation

[1] - ENCE_3501_Lab_4.pdf

[2] - Lab_4_Inverters.pdf